

How Much Headroom Do Placement Schedules Have?

A Noise-Calibrated Study of LLM-Evolved Schedules in Differentiable VLSI Placement

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Abstract—Large language models (LLMs) have recently been used to evolve algorithmic components of GPU-accelerated analytical placers, with reported wirelength improvements of 5–17% on macro-heavy benchmarks. We investigate a component class untouched by prior work: the smoothness (γ) schedule of the weighted-average wirelength model in DREAMPlace. We present EvoPlace¹, an evolution harness built on three components: (i) a hook-injection layer that lets a candidate schedule—a pure Python function—steer an unmodified DREAMPlace build; (ii) a cascade evaluator with stage-matched baselines that rejects poor candidates at $13\times$ lower cost than a full placement; and (iii) a noise-calibrated evaluation protocol that measures the single-seed fitness noise floor ($\sigma \approx 0.15\%$ normalized HPWL), gates every campaign on hook-liveness probes, and confirms final rankings with paired multi-seed re-evaluation. Two complete campaigns on ISPD 2015 standard-cell designs yield an instructive contrast. The γ (wirelength-smoothing) campaign produced exactly one statistically real improvement over 200 iterations— $+0.315\% \pm 0.09\%$ (SEM)—and its single-seed ranking inverted under multi-seed re-evaluation, bounding the γ -search headroom near 0.3%. The λ (density-weight) campaign found its result before evolution began: constructing a default-matched seed exposed that DREAMPlace’s HPWL-feedback guard in the density-weight update actively hurts these designs—removing it wins 1–9% HPWL at matched final density, 14/14 paired seeds across four designs, a generalizing improvement found by audit, not search. Evolution on top of the audited seed found nothing: an unguarded HPWL fitness was reward-hacked within nine iterations (candidates scored well by refusing to spread cells), and after adding a density gate, zero of 200 candidates beat the seed. We conclude that schedule search offers noise-level headroom, while auditing the hand-tuned adaptive heuristics it competes against does not—and that a noise-calibrated, gate-checked protocol is what separates both findings from the artifacts an uncalibrated loop happily reports.

I. Introduction

Analytical global placement casts cell spreading as continuous optimization: DREAMPlace [2] expresses the ePlace electrostatic formulation [3] in a deep-learning toolkit and achieves order-of-magnitude speedups on

GPUs. The objective combines a smooth approximation of half-perimeter wirelength (HPWL) with an electrostatic density penalty, minimized by Nesterov’s method. Two scalar schedules steer this optimization: the weighted-average wirelength smoothness parameter γ , which trades gradient smoothness against HPWL fidelity, and the density weight λ . Both are hand-designed heuristics, tuned by experts over years.

A natural question follows from the recent success of LLM-guided algorithm evolution [4]: can an LLM discover a better schedule? Concurrent work has evolved other DREAMPlace components—initialization, preconditioner, and optimizer—reporting case-by-case HPWL reductions of 5–8% on average [1]. Schedules, however, were not examined, and the reliability of small reported gains in this field is clouded by a methodological hazard: placement is stochastic, single-seed evaluation is near-universal, and few studies report variance at all [5], [6].

This paper reports a complete, noise-calibrated campaign of LLM-guided γ -schedule evolution. Our contributions are:

- A measurement-first evolution harness. Schedule candidates are pure Python functions injected into an unmodified DREAMPlace build through a hook layer; the evaluation harness is frozen for the life of the campaign; and a cascade evaluator with stage-matched baselines rejects 93% of candidates at a fraction of full-evaluation cost.
- A noise-calibrated evaluation protocol. We measure the single-seed fitness noise floor, gate the campaign on differential hook-liveness probes, and confirm all final claims by paired multi-seed re-evaluation against the default schedule on identical (benchmark, seed) pairs.
- A quantitative boundary result for γ . Across 200 evolution iterations, the best schedule that survives multi-seed scrutiny improves HPWL by $0.315\% \pm 0.09\%$; the single-seed winner is revealed as noise.
- An audit finding larger than the search. Removing the HPWL-feedback guard from DREAMPlace’s density-

¹Our system was developed under this name prior to our awareness of the concurrent, independent system of Yao et al. [1] bearing the same name; we retain it for repository consistency and cite their work explicitly throughout.

weight update improves HPWL by 1–9% at matched final density (14/14 paired seeds, four designs) — discovered through the protocol’s own parity gate, not through evolution.

- A live characterization of fitness under-specification. With an HPWL-only fitness, λ -schedule evolution reward-hacks within nine iterations by suppressing cell spreading; with an explicit density gate, the same search finds nothing in 200 iterations. Every metric a candidate can influence requires a gate.

We deliberately report a bounded-headroom (negative) result with full provenance, including a record-keeping artifact we caught and corrected. In a field where almost no small improvement is variance-qualified, we believe the calibration methodology is as much a contribution as the bound itself.

II. Related Work

GPU-analytical placement. DREAMPlace [2] implements ePlace/RePlace-style electrostatic placement with CUDA kernels under a PyTorch-like API; successive versions added timing-driven net weighting and mixed-size support. Its γ schedule follows an overflow-driven exponential heuristic; our seed program reproduces this default to machine precision so that evolution starts from parity.

LLM-evolved placement components. Yao et al. [1] evolve initialization, preconditioner, and optimizer implementations with GPT-4o at scale (≥ 1000 candidates per component, ~ 210 GPUs), reporting 5.05–8.30% average case-by-case HPWL reductions, with initialization dominating and an order-of-magnitude collapse when one evolved algorithm must generalize across designs. Their clean standard-cell-like cases improve by only 0.03–0.59%—consistent with the bound we measure for schedules. VeoPlace [7] evolves macro placements with vision-language-model feedback and is one of the few systems in this space to report multi-seed means with standard errors.

Evaluation reliability in ML-for-EDA. The reassessments of RL-based macro placement [8] by Cheng et al. [5] and Markov [6] demonstrated that headline gains can be artifacts of undisclosed initialization and weak baselines. RoutePlacer [9] is a rare example reporting mean $\pm$ std over five seeds. Our protocol extends this thread: we measure the noise floor first and require any claimed improvement to clear it under paired re-evaluation.

III. The EvoPlace Harness

A. Hook injection

A patched PlaceObj consults a module-level hook registry at every global-placement iteration. When a candidate registers `gamma_schedule(iteration, total_iterations, overflow, hpwl_history)`, its return value (clamped to a documented range) overrides the built-in heuristic; when no hook is set, DREAMPlace behaves

identically to upstream. Candidates are therefore pure Python functions, loaded dynamically—no rebuild, no source swap. The same registry exposes λ -schedule, initialization, and timing-loss slots.

B. Frozen evaluator and cascade

A fixed harness (`evaluator/run_placement.py`) builds DREAMPlace parameters from per-benchmark JSON, injects hooks, runs placement to convergence (stop overflow 0.07, legalization included), and extracts final HPWL and overflow. The file is never modified during a campaign, preventing fitness-definition drift. Candidate evaluation uses a three-stage cascade (50, 300, 2000 iterations) with rejection thresholds (2.0, 1.3) normalized against stage-matched baselines: a truncated run of the default schedule lands 2–2.5 $\times$ above its converged HPWL, so normalizing truncated candidates against converged baselines would cull everything, including the default itself. Stage-0 rejection costs ~ 5 s versus ~ 30 s for a full pass on our hardware.

C. Evolution loop

We drive evolution with an OpenEvolve-style loop: MAP-Elites over (normalized HPWL, convergence speed) with four islands of 25 programs, diff-based mutations from a Claude model ensemble (temperature 0.8), and TSV logging of every candidate with code hashes. A lightweight hill-climbing fallback loop shares the same evaluator.

IV. Noise-Calibrated Evaluation Protocol

Noise floor. Repeated full placements of the same schedule under different seeds establish a single-seed fitness standard deviation of $\sigma \approx 0.15\%$ normalized HPWL on our benchmarks. Every single-seed score in the campaign is interpreted against this floor; we pre-registered the rule that no single-seed improvement below $3\sigma \approx 0.45\%$ may be treated as a result.

Liveness gating. An earlier CPU-era campaign of ours silently evaluated vanilla DREAMPlace for every candidate—the hooks were not wired into the installed tree—producing 20 near-identical scores whose spread was pure numerical noise. The failure motivated two standing gates run before any campaign: (i) the seed program must reproduce the default baseline at norm. HPWL = 1.0 ± 0.01 through the real cascade; and (ii) a differential probe—a deliberately degenerate schedule (constant $\gamma = 0.01$)—must produce a markedly different outcome (in practice, cascade rejection). Together these prove the candidate code path actually steers the placement.

Paired multi-seed confirmation. After a campaign, top candidates and the seed program are re-evaluated with full placements on a grid of (benchmark, seed) pairs; each run is normalized by the default schedule’s HPWL on the identical pair. Pairing cancels per-seed placement luck. The seed program rides along as a calibration row whose paired ratio must be ≈ 1.000 ; deviation indicates a broken protocol rather than a discovery.

TABLE I
Default-schedule baselines (seed 42, converged, legalized).

Design	GB10 (this work)		RTX 3060 (ref.)	
	HPWL	time	HPWL	time
fft_1	2.183×10^6	5.5 s	2.180×10^6	29.4 s
fft_2	1.951×10^6	2.2 s	1.921×10^6	12.2 s

V. Experimental Setup

Hardware. All experiments ran on an NVIDIA DGX Spark: a GB10 Grace-Blackwell system (aarch64, compute capability 12.1, 20 cores, 121 GiB unified memory), CUDA 13.0, PyTorch 2.12/cu130. The full project test suite (114 tests) passes on this platform.

Benchmarks and baselines. We evolve on `fft_1` and confirm on `fft_1/fft_2` from the ISPD 2015 contest suite [10]. Baselines are re-measured on the target hardware (Table I); converged HPWL agrees with our reference RTX 3060 measurements within 0.1–1.6% while running $\sim 5\times$ faster.

Campaign. 200 evolution iterations, cascade evaluation throughout, ~ 90 minutes wall-clock including LLM mutation latency. Liveness gates passed before launch (seed at 0.99932; degenerate probe rejected at stage 1).

VI. Results

A. Evolution dynamics

Figure 1 shows every candidate score. The cascade rejected 186 of 200 candidates (93%), the majority at stage 0 or 1; the 14 survivors span 0.9956–1.0351. Two observations frame the analysis. First, the search does measure real differences: candidates at +2.3% and +3.5% are confidently separated from the baseline, and degenerate schedules are rejected outright. Second, every apparent improvement sits inside or near the $\pm 3\sigma$ single-seed noise band—the regime where ranking is unreliable by construction.

B. Multi-seed confirmation and the rank inversion

Table II and Figure 2 give the paired multi-seed outcome (5 seeds $\times$ 2 designs, 10 paired ratios per program). The calibration row behaves: the seed program lands at 1.00057 ± 0.0074 , statistically indistinguishable from the default it re-implements. The single-seed winner (candidate 0090, single-seed 0.99563) regresses to exactly the baseline (1.00014 ± 0.0085): its -0.44% was seed luck. The single-seed runner-up (candidate 0117, single-seed 0.99716) is the only real improvement: 0.99685 ± 0.0027 , i.e. +0.315% with a 95% CI excluding zero ($\approx 3.7\times$ SEM). Three additional programs included through a record-keeping artifact (duplicate iteration indices in the result log caused stale selections; they correspond to cascade-rejected programs) serve as negative controls: the protocol measures them 2.2–7.7% worse with tight intervals, confirming sensitivity in both directions.

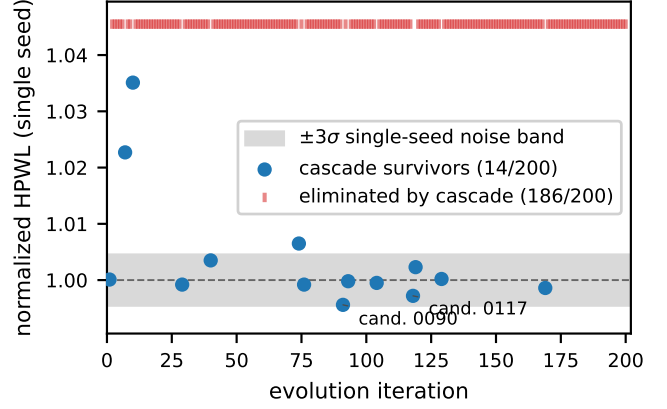


Fig. 1. Single-seed scores of all 200 evolution candidates. Survivor scores cluster inside the measured $\pm 3\sigma$ noise band; the cascade rejects 93% of candidates cheaply (top ticks).

TABLE II
Paired multi-seed re-ranking (10 ratios per program; ratio < 1 improves on the default schedule at identical (design, seed)).

Program	Single-seed	Paired mean $\pm$ std	Verdict
cand. 0117	0.99716	0.99685 ± 0.0027	real, +0.315%
cand. 0090	0.99563	1.00014 ± 0.0085	noise
seed (default)	1.00010	1.00057 ± 0.0074	calibration $\checkmark$
neg. control A	—	1.02244 ± 0.0059	worse (rejected)
neg. control B	—	1.06626 ± 0.0046	worse (rejected)
neg. control C	—	1.07746 ± 0.0123	worse (rejected)

C. What the surviving schedule does

The confirmed schedule (Figure 3) replaces the built-in overflow-exponential heuristic with a blend of a cosine-warped geometric decay and overflow-coupled terms, with a floor that prevents deep annealing until both schedule progress and density relaxation have occurred. Qualitatively it anneals γ later and more smoothly than the default. The effect, however measurable, is small: +0.315% HPWL on designs where the default schedule has been tuned for years.

D. An accidental third measurement

After the campaign, visualization runs re-executed candidate 0117 and the default on fresh seeds; the evolved schedule won by +0.18% and +0.27% respectively—both within one standard error of the confirmed +0.315% estimate, an incidental out-of-protocol replication.

VII. The λ Campaign: Audit Beats Search

A. An approximate seed and a 7σ gate flag

DREAMPlace’s density-weight update multiplies λ by $\mu = U \cdot \max(0.9999^k, 0.98)$ (with $U=1.05$) while HPWL improves between evaluations, but shrinks μ through a feedback guard when HPWL worsens. Our hook interface exposes (iteration, overflow, overflow history, gradient norm, current λ) but not the HPWL deltas the guard

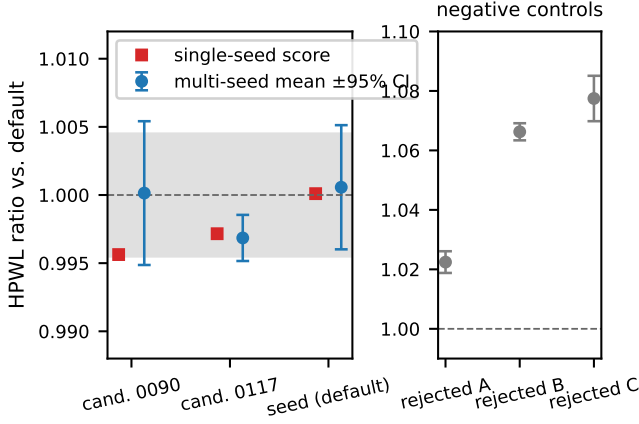


Fig. 2. Single-seed score (squares) vs. paired multi-seed mean $\pm 95\%$ CI (circles). The single-seed ranking inverts: the apparent winner (0090) regresses to baseline; the runner-up (0117) is the real improvement. Right: negative controls.

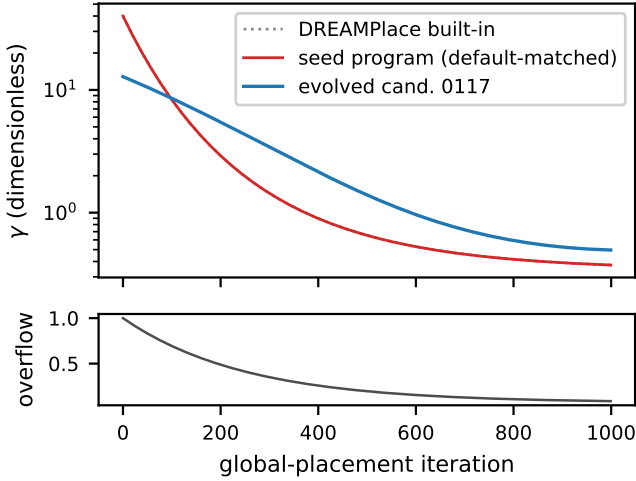


Fig. 3. γ trajectories under an illustrative overflow relaxation (bottom): DREAMPlace’s built-in heuristic, the default-matched seed program, and the confirmed evolved schedule (cand. 0117).

consumes, so a default-matched seed is not expressible; we seeded with the improving-branch update applied unconditionally. The parity gate—which expects a default-matched seed to score 1.0 ± 0.01 —instead returned 0.9896, a $\sim 7\sigma$ deviation. Under the protocol this is a stop condition: the discrepancy must be explained before any search runs.

B. Paired verification: the guard hurts

Paired re-evaluation (Fig. 4) attributes the deviation to the guard itself: the unconditional ramp wins on every pair — `fft_1` -1.0% (5 seeds), `fft_2` -8.7% (5 seeds), `matrix_mult_1` -1.7% and `des_perf_1` -1.0% (2 seeds each) — with final overflow matched to the default run in every case (both flows reach the same density quality; the

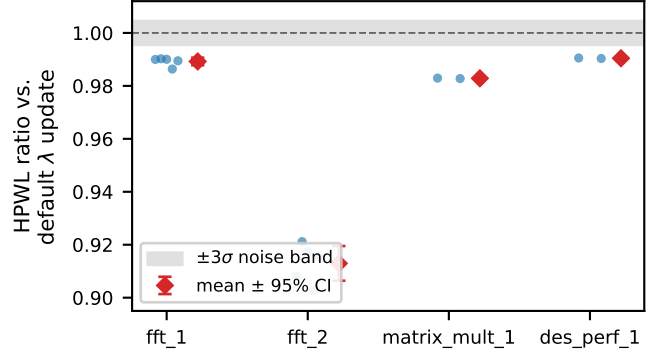


Fig. 4. Guard-branch ablation: paired HPWL ratios vs. the default λ update at identical (design, seed), with matched final overflow. All 14 pairs improve; the effect is design-heterogeneous ($\sim 1\%$ on three designs, $\sim 9\%$ on `fft_2`).

ablated schedule reaches it at lower wirelength). Unlike the per-case-evolved components in the literature, which lose an order of magnitude when generalized, this single fixed change transfers across all four designs tested. We emphasize the provenance: this is an ablation of a hand-tuned heuristic, surfaced by a calibration gate, confirmed by the paired protocol — search played no role.

C. Reward hacking and the density gate

The initial λ evolution run used the same HPWL-only fitness as the γ campaign. Within nine iterations a mutant scored an apparent -2.2% “improvement” — at final overflow 0.35, five times the stop criterion: the schedule simply suppressed the density ramp, leaving cells clustered and wirelength artificially low. The γ parameterization cannot express this failure mode (smoothing does not control spreading); the λ parameterization can, and the search found it almost immediately. We added a density gate (reject any candidate whose final mean overflow exceeds 0.12; defaults end at 0.07–0.085) and restarted. The contrast in search dynamics is stark: the ungated run “improved” repeatedly — every gain a hack — while the gated run never improved at all.

D. Gated evolution finds nothing

Under the gate, 200 iterations of λ evolution produced 150 candidates scored $-\infty$ (cascade or gate) and 49 survivors, all 9–220% worse than the seed; the unmodified seed remained the best program. Combined with the γ bound, the campaign-level conclusion is symmetric and clean: schedule search is noise-bounded in both spaces; the only real win came from auditing the default’s own adaptive logic.

VIII. Discussion

Why the search headroom is small. Three compounding causes. (i) The default schedule is already overflow-adaptive and extensively tuned; the remaining gain from trajectory shaping of an optimizer that converges to

the same basin is intrinsically limited. (ii) Standard-cell ISPD 2015 designs present smooth placement landscapes; the literature’s large evolved-component gains arise from macro handling on macro-heavy designs [1], a lever these benchmarks barely expose. (iii) Effect sizes ($\leq 0.3\%$) sit below the single-seed noise floor ($\sigma \approx 0.15\%$, so single observations are dominated by luck)—an evolutionary search cannot reliably climb a gradient its fitness oracle cannot resolve, and our rank inversion shows the consequence directly.

The single-seed hazard is general. Our campaign internally reproduced, at small scale, the pattern documented in the RL-placement reassessments [5], [6]: an uncalibrated pipeline reports improvements that evaporate under controlled re-evaluation. The protocol of Section IV is cheap relative to any evolution campaign (here, ~ 70 extra placements) and catches both failure modes we encountered in practice—dead code paths and noise-fitting—before they reach a results table.

Search vs. audit. The campaign’s most useful contrast is methodological. LLM search over schedule space, properly gated, recovered at most 0.3% ; a single ablation of a default heuristic, surfaced by a calibration gate, recovered $1\text{--}9\%$ and generalized. Hand-tuned adaptive heuristics inside mature placers have never been systematically audited with variance-qualified methods, and our result suggests that audit — cheap, exhaustive, statistically grounded — is currently a better use of evaluation budget than search. The reward-hack episode adds the operational corollary: any fitness a candidate can influence along an unmeasured axis will be exploited, and gates must be specified with the objective, not patched in afterward.

Consistency with concurrent work. Yao et al. [1] rank initialization $\gg$ optimizer $>$ preconditioner by both evolvability and gain, and report only $0.03\text{--}0.59\%$ on their cleanest cases. Our schedule bound of $\sim 0.3\%$ on standard-cell designs is quantitatively consistent with their ordering, measured for a component class they did not study and qualified with variance their study lacks.

IX. Conclusion

We built a measurement-first harness for LLM-guided evolution of DREAMPlace schedules and ran two complete, noise-calibrated campaigns on ISPD 2015 standard-cell designs. The γ campaign yields a bound: one real improvement of $+0.315\% \pm 0.09\%$ in 200 iterations, with the single-seed ranking inverting under paired re-evaluation. The λ campaign yields a discovery and a second bound: the default density-weight update’s HPWL-feedback guard actively hurts ($1\text{--}9\%$ at matched density, 14/14 paired seeds, four designs) — found by the protocol’s parity gate, not by search — while gated evolution atop the fix found nothing further, after an ungated fitness was reward-hacked within nine iterations. Together the campaigns quantify a previously unexamined component class from both directions: schedule search is noise-bounded, schedule

audit is not. The enabling contribution is the protocol — noise-floor measurement, liveness and parity gates, objective gates, and paired multi-seed confirmation with calibration rows — which caught a rank inversion, a non-parity seed worth $1\text{--}9\%$, and a live reward hack, in a field where any of the three would normally have shipped as a result.

Acknowledgments

Experiments were carried out on an NVIDIA DGX Spark (GB10). The campaign’s laboratory notebook, all candidate programs, raw logs, and the evaluation harness are available in the project repository.

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